

The cubic size bound: current proof state

the proved chain, the single open lemma, and exactly why it resists

Secretary working note — all numeric facts machine-validated

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1 Symbols (every function defined once, recursively)

Regexes: $\mathbf{0}, \mathbf{1}$, char c , sequence $r_1 \cdot r_2$, alternation $\sum_i r_i$, star r^* . An *atom* a is a char, alternation or star (non-., non-unit).

$|r|$ (size)

$$|\mathbf{0}| = |\mathbf{1}| = |c| = 1, \quad |r_1 r_2| = 1 + |r_1| + |r_2|, \quad |\sum r_i| = 1 + \sum |r_i|, \quad |r^*| = 1 + |r|.$$

$\|X\|$ (set size)

$\sum_{x \in X} |x|$ over the *distinct* members of a finite set/list X .

S (strong normaliser; “rsimpStrong_raw”)

simplify bottom-up, plugging sequences with σ_7 (so the guarded $a^* \cdot a^* \rightarrow a^*$ fires at every level), and idempotent-collapse stars:

$$S\mathbf{0} = \mathbf{0}, \quad S\mathbf{1} = \mathbf{1}, \quad Sc = c, \quad S(r_1 r_2) = \sigma_7(Sr_1, Sr_2), \quad S(\sum r_i) = A(\text{flat}(\text{map } S r_i)),$$

$$S(r^*) = (\text{case } Sr \text{ of } \mathbf{0} \Rightarrow \mathbf{1} \mid \mathbf{1} \Rightarrow \mathbf{1} \mid s^* \Rightarrow s^* \mid s \Rightarrow s^*)$$

(A flattens, dedupes and rebuilds an alternation). The single collapse $a^* \cdot a^* \rightarrow a^*$ — realised by σ_7 in the sequence case, recursively — is the source of all the difficulty below.

$\sigma(h, k)$ (weak plug: “ h then k ”)

does all sequencing; **never** collapses $a^* \cdot a^*$:

$$\sigma(\mathbf{0}, k) = \mathbf{0}, \quad \sigma(\mathbf{1}, k) = k, \quad \sigma(r_1 r_2, k) = \sigma(r_1, \sigma(r_2, k)), \quad \sigma(a, \mathbf{0}) = \mathbf{0}, \quad \sigma(a, \mathbf{1}) = a, \quad \sigma(a, k) = a \cdot k.$$

$\sigma(p, \mathbf{1})$ *canonicalises* p alone (right-associate, erase units, absorb $\mathbf{0}$; e.g. $\sigma((ab)c, \mathbf{1}) = a(bc)$, $\sigma(a\mathbf{1}, \mathbf{1}) = a$).

$\sigma_7(h, k)$ (strong plug: “rsimp7_SEQ_atom”)

σ *plus* the guarded $a^* \cdot a^* \rightarrow a^*$ collapse applied **only at the top head**: $\sigma_7(a^*, a^*) = a^*$, $\sigma_7(a^*, a^* \cdot k) = a^* \cdot k$, and $\sigma_7(h, k) = \sigma(h, k)$ otherwise. So σ_7 sits between σ (no collapse) and S (collapses everywhere) — it collapses just the leading star pair.

∂k (frontier)

split the top alternation of k into its branches ($\partial \mathbf{0} = \emptyset$, $\partial(\sum r_i) = \bigcup \partial r_i$, else $\{k\}$).

$\text{dl}(x)$ (open one regex to linear forms; “row_dlforms”)

$$\text{dl}(\mathbf{0}) = \emptyset, \quad \text{dl}(\sum r_i) = \bigcup_i \text{dl}(r_i), \quad \text{dl}((\sum p_i) \cdot k) = \bigcup_{p \in p_i} \text{dl}(\sigma_7(p, k)), \quad \text{dl}(r) = \partial r \text{ (otherwise)}.$$

Distribute a top alternation, and an alternation-headed sequence’s branches over the tail via σ_7 ; bottom out at the frontier. (This is the *only* other place $a^* \cdot a^*$ can collapse — via σ_7 in the $\cdot$ case.)

$o(x)$, $f(k)$ (opened size of one regex / of a frontier)

$o(x) := \|\text{dl}(Sx)\|$ (strong-simplify, open, dedupe, sum sizes); $f(k) := \|\bigcup_{q \in \partial k} \text{dl}(Sq)\|$.

$D_c(r)$ (Antimirov derivative list)

the partial-derivative “rows”, sequencing by σ :

$$D_c \mathbf{0} = D_c \mathbf{1} = [], \quad D_c(\text{char } d) = [\mathbf{1} \text{ if } d=c \text{ else }], \quad D_c(\sum rs) = \bigoplus_i D_c(r_i),$$

$$D_c(r_1 r_2) = [\sigma(p, r_2) \mid p \in D_c(r_1)] \oplus (D_c(r_2) \text{ if } r_1 \text{ nullable}), \quad D_c(r^*) = [\sigma(p, r^*) \mid p \in D_c(r)].$$

$\widehat{D}_c(r)$ (canonicalised rows)

$:= [\sigma(p, \mathbf{1}) \mid p \in D_c(r)]$ — every row put in canonical form so the dedup below actually merges associativity/unit variants (keeping the row set small).

$F(r, s)$ (normalised front; “afactored1”)

read the whole string s with *no S*: starting from $[r]$, apply for each letter c the normalised step $\nu_c(\rho) := \text{dedup}(\text{flat } \bigcup_{q \in \rho} \widehat{D}_c(q))$,

$$F(r, \varepsilon) = [r], \quad F(r, c:s) = F(\text{root of } \nu_c([r]) \text{ etc.}) = \nu_{c_n} \circ \dots \circ \nu_{c_1}([r]).$$

$\Delta_c(\rho)$ (one strong step; “rpder_strong_rows_raw”)

derive+canonicalise each row, strong-simplify with S , flatten, strong-prune with P , flatten, dedupe:

$$\Delta_c(\rho) := \text{dedup}\left(\text{flat } P\left(\text{flat } \bigcup_{q \in \rho} [S(x) \mid x \in \widehat{D}_c(q)]\right)\right).$$

$P(\rho)$ (strong prune; “rsimpStrong_prune_rows_raw”)

scan ρ left-to-right keeping the pruned rows seen so far, replacing each row by its pruning against all earlier ones (rows are *shrunk*, *never dropped*). The only non-trivial pairwise rule fires on two rows *sharing a right factor* k : $((\sum lrs) \cdot k, (\sum rrs) \cdot k) \mapsto \sigma_\tau(\sum (rrs \setminus lrs), k)$ — delete from the later row’s head-alternation the branches already present in the earlier one $(rrs \setminus lrs)$; every other pair leaves the later row unchanged. Language-preserving and size-non-increasing.

$\Theta(\rho)$ (open a row list; “row_dlformss”)

$$\Theta(\rho) := \bigcup_{q \in \rho} \text{dl}(q).$$

$p(r, k)$ (potential)

an over-approximation of the opened size of r ’s derivative universe against k , defined to recurse on r and bound it from above ($\|\mathbf{U}(r)\| \leq p(r, \mathbf{1})$, proved):

$$p(\mathbf{0}, k) = 1, \quad p(\mathbf{1}, k) = 1 + o(k) + f(k), \quad p(\text{char } c, k) = o(\sigma(c, k)) + 1 + o(k) + f(k),$$

$$p(r_1 r_2, k) = p(r_1, \sigma(r_2, k)) + p(r_2, k), \quad p(\sum rs, k) = 1 + o(\sigma(\sum rs, k)) + \sum_q p(q, k),$$

$$p(r^*, k) = \underbrace{o(\sigma(r^*, k))}_{\text{head}} + \underbrace{p(r, \sigma(r^*, k))}_{\text{body: } r \text{ against its own star}}.$$

$\mathbf{U}(r)$ (strong-opened universe; “strong_opened_live_row_universe”)

open (via $\text{dl} \circ S$) every regex of r ’s derivative universe $\mathcal{D}(r)$:

$$\mathbf{U}(r) := \bigcup_{p \in \mathcal{D}(r)} \text{dl}(Sp), \quad \mathcal{D}(r) := \{\mathbf{0}, \mathbf{1}, r\} \cup \text{paths}(r) \cup \partial r \cup \bigcup_{q \in \text{paths}(r)} \partial q,$$

where $\text{paths}(r)$ is the (finite) set of continuations reachable by repeatedly deriving r . Proved static facts: $\#\mathcal{D}(r) \leq |r| + 2$ (linear) and each member has size $\leq (|r| + 2)^2$ (quadratic) — the linear $\times$ quadratic that the cubic budget rests on. $\|\mathbf{U}(r)\|$ is what the Gate must bound.

$\tau(r)$ (trace), w_k (event cost)

the explicit event-list expansion of $p(r^*, \cdot)$, with the continuation tracked as a prefix stack xs that ap instantiates:

$$ap(a, [], k) = \sigma(a^*, k), \quad ap(a, x:xs, k) = \sigma(x, ap(a, xs, k)).$$

Events are AU (unit), AO xs (open), AF xs (frontier), with cost $w_k(\text{AU}) = 1$, $w_k(\text{AO } xs) = o(ap(r, xs, k))$, $w_k(\text{AF } xs) = \|\bigcup_{q \in \partial(ap(r, xs, k))} dl(Sq)\|$. The trace at mirrors p , pushing the right part onto the stack:

$$at(\mathbf{0}, xs) = [\text{AU}], \quad at(\mathbf{1}, xs) = [\text{AU}, \text{AO } xs, \text{AF } xs], \quad at(c, xs) = \text{AO}(c:xs) \# [\text{AU}, \text{AO } xs, \text{AF } xs],$$

$$at(\sum rs, xs) = \text{AU} \# \text{AO}(\sum rs:xs) \# \bigoplus_q at(q, xs), \quad at(pq, xs) = at(p, q:xs) \oplus at(q, xs), \quad at(r^*, xs) = \text{AO}(r$$

$$\tau(r) := \text{AO } [] \# at(r, []), \text{ with the proved identity } p(r^*, k) = \sum_{e \in \tau(r)} w_k(e).$$

minor list operations

flat flattens nested alternations and drops $\mathbf{0}$ s; dedup keeps the first occurrence of each row; $\oplus/\bigoplus$ is list append/concat; “ r nullable” means r matches ε ; $rrs \setminus lrs$ drops from rrs the entries already occurring in lrs .

2 The goal, and why it is cubic

In the clean fragment (classical regexes, normal form, no bounded repetition): read s , take *one* strong step on the next letter c , open the resulting rows and sum their sizes. The thesis target is that this is cubic in $|r|$, uniformly in s :

$$\boxed{\|\Theta(\Delta_c(F(r, s)))\| \leq 2(|r| + 3)^3} \quad (\text{Gate})$$

Why: a cubic ceiling independent of $|s|$ makes the lexer’s space polynomial in the regex regardless of input length; it is the crux. *Design note*: $F(r, s)$ is the cheap $\mathbf{S}$ -free accumulation (the D law and the static cubic front are proved on it); the $\mathbf{S}$ collapse $a^* \cdot a^* \rightarrow a^*$ and the pruning are applied *once*, only in the final Δ_c . So $\mathbf{S}$ ’s collapse — and the global cancellation it induces — live only at that last step, and that is the wall (§4–§5).

3 The proved chain (top to bottom)

All machine-checked, no **sorry**. Each line is justified by the one below it.

1. **[proved] Gate $\Leftarrow$ root potential.** The next-step opened rows lie inside $U(r)$ (monotonicity), and $\|U(r)\| \leq p(r, \mathbf{1})$ is proved, so the Gate follows from $p(r, \mathbf{1}) \leq 2(|r| + 3)^3$. *Why*: trade a statement about an opened *set* for a recursively-defined numeric *upper bound* p that recurses cleanly on the regex.
2. **[proved] The cube-shell.** For all r, k : $p(r, k) \leq (|r| + |k|)^3 - (|k|)^3$ (CS); at $k = \mathbf{1}$, $p(r, \mathbf{1}) \leq (|r| + 1)^3 - 1 \leq 2(|r| + 3)^3$, closing step 1. *Why*: (CS) is a difference of cubes — it telescopes additively, hence is provable by induction on r .
3. **[proved] (except one case) The cube-shell induction (on r).** $\mathbf{0}, \mathbf{1}$, char are leaves; $\sum$ and $r_1 r_2$ compose the children’s shells — *all proved*. The one open constructor is r^* . *Why*: the r^* recursion feeds the body r against an *inflated* continuation (step 5), so the naive use of (CS) as the body’s hypothesis overshoots by $\approx 7\times$ and cannot close that case.

4. **[proved] r^* case $\Leftarrow$ the affine envelope.** With $M := |r^*| = 1 + |r|$, it suffices to prove

$$\mathbf{p}(r^*, k) \leq M^3 + 3M^2 |k|. \quad (\text{Env})$$

Indeed $M^3 + 3M^2 |k| \leq (M + |k|)^3 - (|k|)^3$ (leftover $3M(|k|)^2 \geq 0$), so $(\text{Env}) \Rightarrow (\text{CS})$ at r^* . **Why:** (Env) is *linear in $|k|$* , where (CS) is quadratic; (Env) is the tight statement at r^* . It is machine-validated true (0 violations).

5. **[proved] Unfolding r^* .** $\mathbf{p}(r^*, k) = \mathbf{o}(\sigma(r^*, k)) + \mathbf{p}(r, \sigma(r^*, k))$. The head is harmless; the **body** opens r against $\sigma(r^*, k) = r^* \cdot k$, its own star re-entry, with continuation inflated by $\approx M$. **Why:** the heart of the difficulty: the body is opened against a continuation containing a copy of its own star. The opened size does *not* grow with the inflation (the re-entry saturates via $a^* \cdot a^* \rightarrow a^*$, used helpfully), but capturing that is the open problem.
6. **[proved] $(\text{Env}) \Leftarrow$ a per-event certificate.** By the trace identity $\mathbf{p}(r^*, k) = \sum_{e \in \tau(r)} \mathbf{w}_k(e)$. Given $a_1, a_B : \text{events} \rightarrow \mathbb{N}$ with
- $$(\text{affine}) \mathbf{w}_k(e) \leq a_1(e) + a_B(e)(|k| - 1), \quad (\text{slope}) \sum_e a_B(e) \leq 3M^2, \quad (\text{intercept}) \sum_e a_1(e) \leq M^3 + 3M^2,$$
- the certificate proves (Env) . **Why:** splits (Env) into a continuation-slope and a continuation-free intercept, each a single sum.
7. **[proved] Slope solved.** a_B a row/frontier count per event gives $\sum_e a_B(e) \leq 3M^2$. **Why:** k sits at the *tail* of every opened row (behind the single anchor r^*), so it adds size linearly.
8. **[proved] Intercept reduced to a single anchor.** Take $a_1(e) := \mathbf{w}_{\text{char } c}(e)$ for a fixed $\text{char } c$: one fixed tail dominates every size-1 tail per event (so (affine) holds), and then $\sum_e a_1(e) = \mathbf{p}(r^*, \text{char } c)$ by the trace identity. **Why:** dissolves a multiplicity trap (a per-event max over all size-1 tails over-counts); one fixed tail collapses the intercept sum back to one potential value.
9. **[proved] Event count is linear.** $|\tau(q)| \leq 4|q|$ (length of the trace). **Why:** the linear factor: the intercept is *(linearly many events) $\times$ (per-event opened size)*, and we must show this product is cubic.

4 The single open lemma

The entire Gate is proved *modulo one scalar inequality* — the intercept of step 8, the *anchor*:

$$\boxed{\mathbf{p}(r^*, \text{char } c) \leq M^3 + 3M^2} \quad \left(= \sum_{e \in \tau(r)} \mathbf{w}_{\text{char } c}(e) \leq M^3 + 3M^2 \right), \quad M = 1 + |r|. \quad (\star)$$

[OPEN]. Machine-validated true: 0 violations over an *exhaustive* enumeration of all in-regime terms up to $|r| = 9$ ($\approx 8.7 \times 10^5$ cases); worst ratio 0.95 at $M = 2$ ($19 \leq 20$), slacker for larger M (true value $\approx 0.5 \times$ budget). Unfolding r^* , $(\star)$ is the *body saturation* $\mathbf{p}(r, \sigma(r^*, \text{char } c)) \leq M^3 + 3M^2 - \mathbf{o}(\sigma(r^*, \text{char } c))$: the body opened against its own star re-entry, budgeted at the *original* (size-1) tail, not the inflated one.

5 Why $(\star)$ resists — a global cancellation

$(\star)$ is $\sum_{\text{events}} (\text{opened size}) \leq M^3 + 3M^2$, with the event count linear (step 9) and each opened size $\leq$ quadratic. Naively “linear $\times$ quadratic = cubic” — yet every *structural* bound (holding pointwise at each event/node/row) overshoots, while the true sum is $\approx 0.5 \times$ the budget:

candidate bound	ratio to budget	fate
true value $\sum_e w_c(e)$	≈ 0.5	what we must bound
crude (events $\times$ max event cost)	$\approx 1.5\times$	overshoots
saturated per-event cap $(M+1)M \times 4 r $	$\approx 4\times$	overshoots
generic quadratic opening $\text{Suc}(sz) \cdot sz$	$\approx 7\times$	overshoots
tight body invariant $(q +1)(M+1)M$	1.0 pointwise	does <i>not</i> telescope ($\sum: 1 + \text{cap} \cdot rs \leq \text{cap}$ false)
deduped universe $\ \text{rows}(r^*)\ $	≈ 0.2 (cubic!)	per-row multiplicity reaches 18; no injection

Diagnosis. The sum is small only by a **global cancellation**: events opening *many* rows have *small* per-row cost; events with *few* rows have *large* cost; high-multiplicity rows are small. Size and multiplicity are *anti-correlated*. A structural induction must bound the worst case *at every event at once*, so it charges high multiplicity and high size simultaneously and overshoots. **No rsize/count structural-induction invariant sees this cancellation.** (Hence the clean “deduped universe is cubic”, ratio 0.2, does not help: the trace charges some rows $18\times$, and the multiplicity-weighted sum has no injection into the deduped set.) On exactly $(\star)$: 8+ external rounds, a 4-agent design panel (a confident *false* bound, caught), and three formalisation attempts — all structural, all overshoot.

6 The shape of a correct proof: the amortised precedent

The sibling problem — the row *count* being linear (the “D law”) — had the same flavour and *was* cracked, not by structural induction but by a **simultaneous induction on two mutually-supporting invariants**: a boundary-strengthened bound T and a strict credit S (with acc the accumulator, $\text{zw}(r) = \#\text{letters} + \#\text{stars} \leq |r|$):

$$T : \#((\partial\sigma(r, k) \cup \text{acc}(r, k)) \setminus \partial k) \leq \text{zw}(r), \quad S : \text{zw}(r) > 0 \Rightarrow 1 + \#(\text{acc}(r, k) \setminus \partial k) \leq \text{zw}(r).$$

S is the amortisation: a constructor may add a boundary row no *local* cap can absorb, but it is paid by the strict spare credit carried in the strengthened T , and $T \wedge S$ (one induction) gives the count bound. *Why*: a global “adds-but-pays” cancellation is captured by carrying a credit in a strengthened invariant, not by a pointwise bound.

The analogue we need (open). A *size* version of $T+S$: a potential Φ on the state (root r^* , sub-term q , bounda plus a strict *size*-credit, so the per-event opened size is paid by the drop of Φ across the trace, telescoping to $\leq M^3 + 3M^2$ at the root, and surviving (i) the $\sum$ case (where pointwise caps fail) by spending the credit once, and (ii) the star re-entry by the boundary being idempotent at the anchor. No concrete such Φ has yet been exhibited.

7 Where the proof can go

1. **Build the amortised Φ (§6) concretely** and verify it telescopes and stays $\leq M^3 + 3M^2$ — the direct analogue of the (successful) count proof; the most promising untried route.
2. **Rethink the opening/p definitions** so the cancellation becomes a *local* invariant (then a structural induction suffices) — it may be the honest answer if no Φ on the current p is local.
3. **Treat $(\star)$ as a validated lemma** — exhaustively checked ($0/8.7 \times 10^5$); the rest of the Gate is fully proved on top of it.

One-line status. The clean-fragment cubic Gate is proved *except* the single scalar $p(r^*, \text{char } c) \leq M^3 + 3M^2$ — true, exhaustively validated, provably beyond structural induction because it encodes a global size $\times$ multiplicity cancellation; the route that fits is the size analogue of the $T+S$ amortised argument that already cracked the row count.